



SPI-DDH

Strategic Partners'
Initiative for
**Data and
Digital Health**



**World Health
Organization**

European Region

Strategic Partners' Initiative for Data and Digital Health

Progress report and
working group proposals



**World Health
Organization**

European Region

Strategic Partners' Initiative for Data and Digital Health

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Abstract:

This report summarizes the work undertaken by the members of the Strategic Partners' Initiative for Data and Digital Health (SPI-DDH) during the period since its inauguration on 13 June 2024 to and including the end of 2024. The report provides high-level detail of the discussion and considerations made by the four SPI-DDH working groups and proposed outputs of each group to be completed during 2025.

KEYWORDS:

DIGITAL HEALTH; DIGITAL TECHNOLOGY; TELEMEDICINE; MEDICAL INFORMATICS; HEALTH INFORMATION SYSTEMS

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Designed by: Kristine Onarheim

Contents

Background	1
Rationale for SPI-DDH to accelerate digital transformation of health systems and services in the WHO European Region	1
Participants in SPI-DDH	1
Process of SPI-DDH working groups	3
Operation	3
Proposals	3
Working group 1. Unlocking more capacity and access to health care with digitalization and responsible AI	3
Working group 2. Leveraging data and digital technologies to bring health care and prevention into the home	4
Working group 3. An ecosystem approach to standards and interoperability in health care	6
Public health working group. Addressing health care workforce mental health through a data-driven, digital lens for resilience and operational efficiency	8
Looking ahead – on track to deliver for countries	10
Aligning the work of SPI-DDH with other key international initiatives for data and digital health	10
References	11

Background

The Strategic Partners Initiative for Data and Digital Health (SPI-DDH) (1) was launched in June 2024 in response to WHO Regional Committee for Europe resolution EUR/RC72/R2 on *leveraging digital transformation for better health in Europe: regional digital health action plan for the WHO European Region 2023–2030* (2), which called for the WHO Regional Office for Europe to provide stewardship for a new “strategic partnership council for data and digital health to share information and encourage collaborative innovation in the European digital health ecosystem”. In response, SPI-DDH was created to bring together a diverse constellation of public and private stakeholders and subject matter experts to tackle key issues relating to the digital transformation of health systems, and to pursue the goal of all people benefitting from safe, affordable and person-centred digital health.

Rationale for SPI-DDH to accelerate digital transformation of health systems and services in the WHO European Region

Digital tools such as mobile applications, telehealth and artificial intelligence (AI) can revolutionize health care, offering new opportunities for the treatment, management and monitoring of health conditions. However, deploying new technologies and managing large volumes of health data held digitally within health systems pose a mix of challenges for countries in Europe, including:

- difficulties in facilitating digital innovation while maintaining data privacy;
- fragmentation of digital health initiatives and approaches;
- persistent barriers to the interoperability of health data held digitally within health systems;
- measuring the quality and impact of investments in digital health; and
- having appropriate regulatory measures to support the scaling of digital solutions in the health system and ensure their safety.

Effective national policy actions enabling the successful transition to digital health services largely depend upon how regional stakeholders organize and arrange services, infrastructure, data exchange, governance and financing together, and understand the inherent interdependency of policy and localities. Upon this basis, SPI-DDH was launched on 13 June 2024 as a multistakeholder platform exploring a handful of priority topics relevant to the digital transformation of health systems and services in Europe. Guided by the needs of health workers and patients and underpinned by the public health values of equity, solidarity, and gender and human rights, the members of SPI-DDH have begun a year-long process to reach consensus on securing a mutual understanding of the drivers of the key issues and developing concise and implementable solutions that contribute directly to achieving the goal of universal health coverage and creating an inclusive future of digital technologies enabling health and well-being for all. Fig. 1 illustrates SPI-DDH’s objectives and structure.

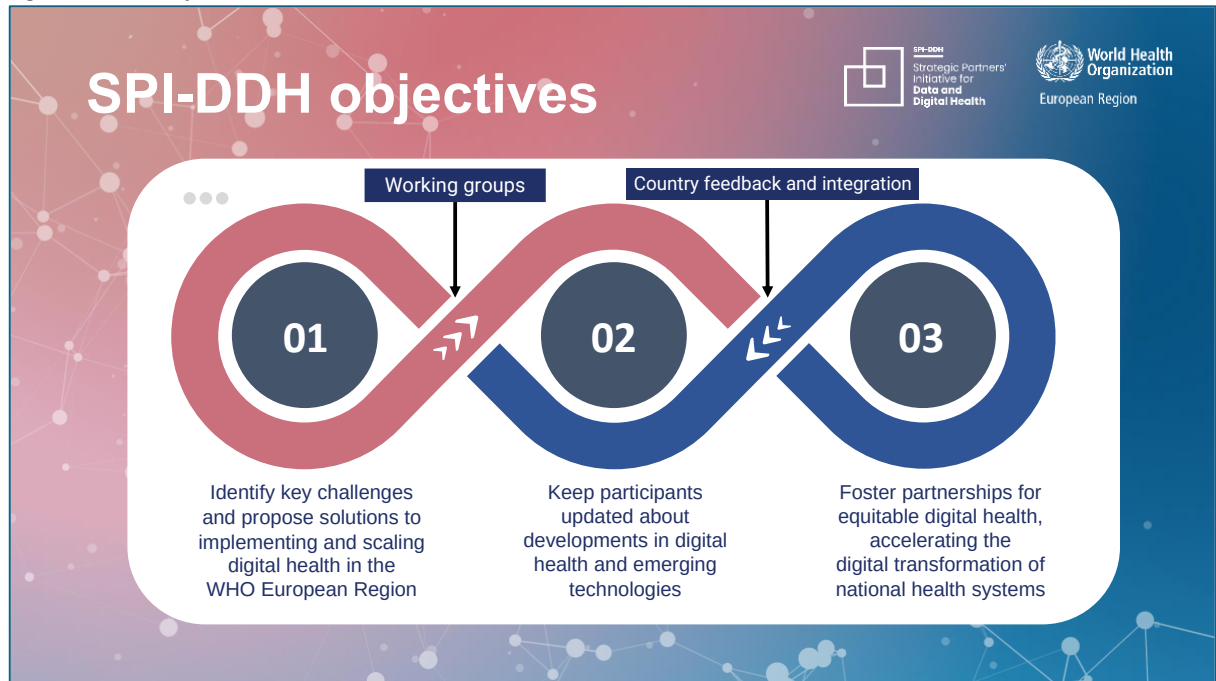
Participants in SPI-DDH

Currently SPI-DDH has the following membership:

- 28 Member States
- 17 private-sector organizations
- 26 nongovernmental organizations

- Three international governmental organizations
- 9 academic institutions
- 16 national institutions
- Five WHO collaborating centres
- Two independent advisors.

Fig. 1. SPI-DDH objectives and structure



Process of SPI-DDH working groups

Operation

Through a process of prioritization and selection, SPI-DDH members chose four topics to explore and develop solutions to address. Working groups were then constituted with members assigned to a group of their choice. Each of the groups is led by a Member State representative elected as chair, supported by two cochairs and a WHO Secretariat facilitator. The WHO Regional Office for Europe Secretariat provides continuous support by assisting in convening meetings of the working groups, ensuring alignment and information exchange across working groups, and ensuring adherence to SPI-DDH's terms of reference (3). Accordingly, the working groups address the following topics:

1. unlocking more capacity and access to health care with digitalization and responsible AI;
2. leveraging data and digital technologies to bring health care and prevention into the home;
3. an ecosystem approach to standards and interoperability in health care; and
4. digital solutions supporting the mental health of the health workforce (public health).

Proposals

Since June 2024, the SPI-DDH working groups have met regularly to undertake the initial tasks of deconstructing the topics addressed and proposing solutions (outputs) that will benefit Member States. The following summarizes each of the working groups' progress to date.

Working group 1. Unlocking more capacity and access to health care with digitalization and responsible AI

Focus: leveraging real-world evidence for AI in health care and demonstrating its value and effectiveness, particularly from a perspective of regulatory approval of AI

Proposed output: compendium of illustrative case studies

The group's initial discussions sought to identify the most acute barriers to unlocking capacity and access to AI in European health systems and to understand how the group's work could be best positioned to contribute to accelerating AI adoption in countries.

As the group members identified a number of topics as highly important, they used a systematic prioritization effort to narrow the focus to what they considered the most important one: ensuring that the regulatory approval around AI leverages real-world data (RWD; see Box 1). While the group felt that identifying best practices for the management of AI products' life cycle and ensuring inclusive digital health solutions were highly important, the selected topic seem more suited for the composition of the working group and the spirit and mandate of SPI-DDH.

Box 1. Definitions

RWD are data relating to patient health status and/or the delivery of health care routinely collected from a variety of sources. Examples of RWD include data derived from electronic health records, medical claims data, data from product or disease registries and data gathered from other sources (such as digital health technologies) that can inform on health status.

Real-world evidence (RWE) is the clinical evidence about the usage and potential benefits or risks of a medical product derived from analysis of real-world data.

Source: US Food and Drug Administration (4)

The working group considered how addressing real-world evidence (RWE) could accelerate health AI adoption in Europe. RWE is important in accelerating health AI adoption, front-loading potential product development issues, making them more transparent and exposing them to higher scrutiny, thus leading to better AI. A faster adoption of AI should in principle lead to generating new capacity.

However, countries in Europe lack a standardized approach to RWD and RWE. Difficulty in capturing RWD, heterogeneity of RWD sources and differing levels of RWD quality mean that countries are unable to leverage RWD being produced in their environments or convert them into meaningful evidence for improving performance and informing the regulatory process for AI. In addition, regulatory frameworks often still ignore or at best underestimate, the potential of RWD, thus in part missing a valuable feedback loop.

The group therefore created a mechanism to solicit case studies on how RWE can be employed and will work to produce a compendium/report based upon them.

Creating a case study repository that captures diverse experiences and insights – whether from health-care providers, public or private health-care payers, industry partners, government bodies or nonprofit organizations – can accelerate learning and sharing across the ecosystem. AI implementation patterns can be adapted and replicated, making knowledge sharing particularly valuable in this quickly evolving field.

Working group 2. Leveraging data and digital technologies to bring health care and prevention into the home

Focus: health care and prevention in the home

Proposed output: framework model and case studies compendium

In the coming decades, a lack of resources, both human and financial, will inevitably characterize health systems, and the demand for care will continue to rise owing to an ageing population having ever increasing noncommunicable diseases and comorbidities. An increasing body of evidence shows that prevention and public health interventions are most effective in people's community context, and that enabling this transition calls for digitally enabled health and social care services and prevention in the home.

The functional integration of primary, secondary and tertiary health care, along with community and social services, is crucial for successful health management. High-quality health systems are characterized by a central role for primary care in prevention and the management of noncommunicable diseases.

A data-exchange infrastructure for integrated person-centred services should make communication seamless and actionable for all users across organizations and among different information technology systems, devices and software applications (Box 2). It should also facilitate quality monitoring across the health continuum: that is, data related to system, process and outcome indicators. Data are an important part of understanding the impact of health determinants and evaluating health and social intervention programmes and their outcomes.

Box 2. Common threads

"I believe it will also be important to identify common threads across the various working groups and ensure the 'digitisation of health systems' is not made of disconnected building blocks but a nicely intertwined architecture."

Giovanni Monti (Working Group facilitator)

By integrating digitally enabled health and social care services with a comprehensive data exchange infrastructure to the home, health systems can become more resilient, efficient and patient centred, ultimately leading to better health outcomes and quality of life for individuals.

Enabling this transition requires that service models, technology, governance and regulations are adapted to the regional context. This typically applies in both urban and rural settings with their differences and commonalities. In addressing these, the working group can develop targeted data and digital health strategies to improve health outcomes and ensure equitable access to care for all populations.

The working group's proposed output was to develop a framework model and compendium of case studies. A practical, problem-based framework would:

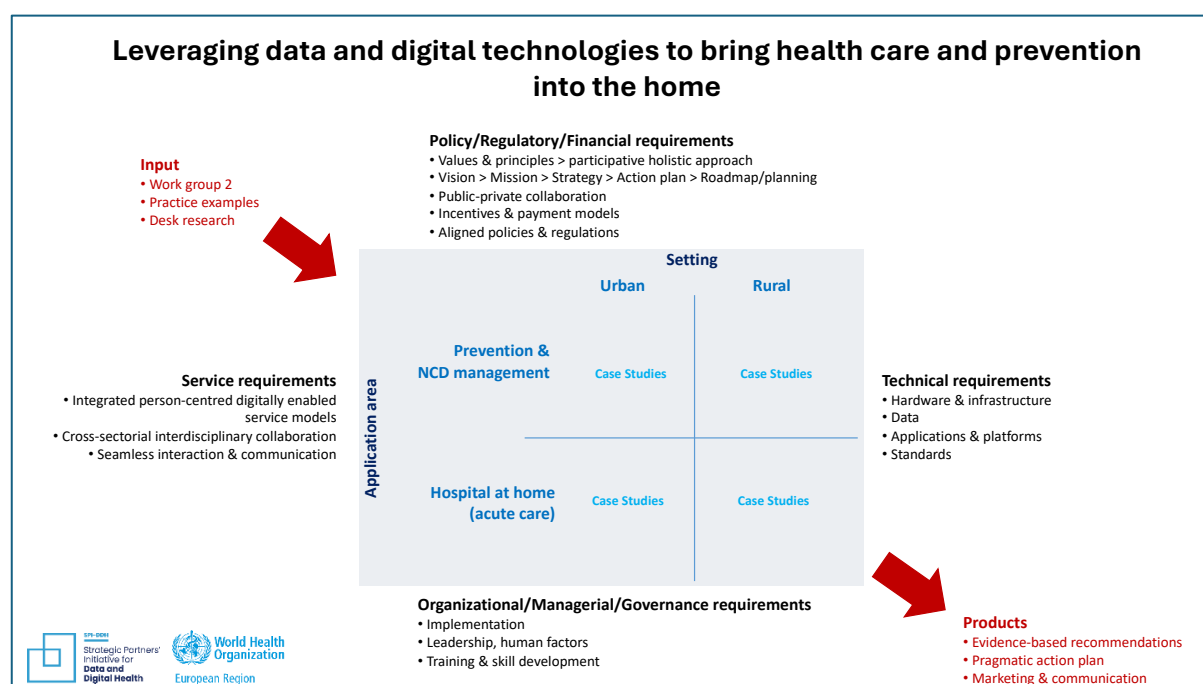
- take users' needs as a starting point;
- address common challenges across different contexts;
- provide practical direction, based on evidence-based recommendations for implementers;
- drive innovation to where it is needed; and
- highlight specific training and skills development needs.

The development process would include:

- undertaking a Delphi survey to define the range of challenges that the framework needs to address;
- gathering use cases/case studies/personas to provide evidence to support recommendations and implementation advice included in framework; and
- identifying and mapping of use cases and refining the framework through multistakeholder cocreation sessions.

Fig. 2 shows a draft adaptation model developed by the working group.

Fig. 2. Draft adaptation model



Working group 3. An ecosystem approach to standards and interoperability in health care

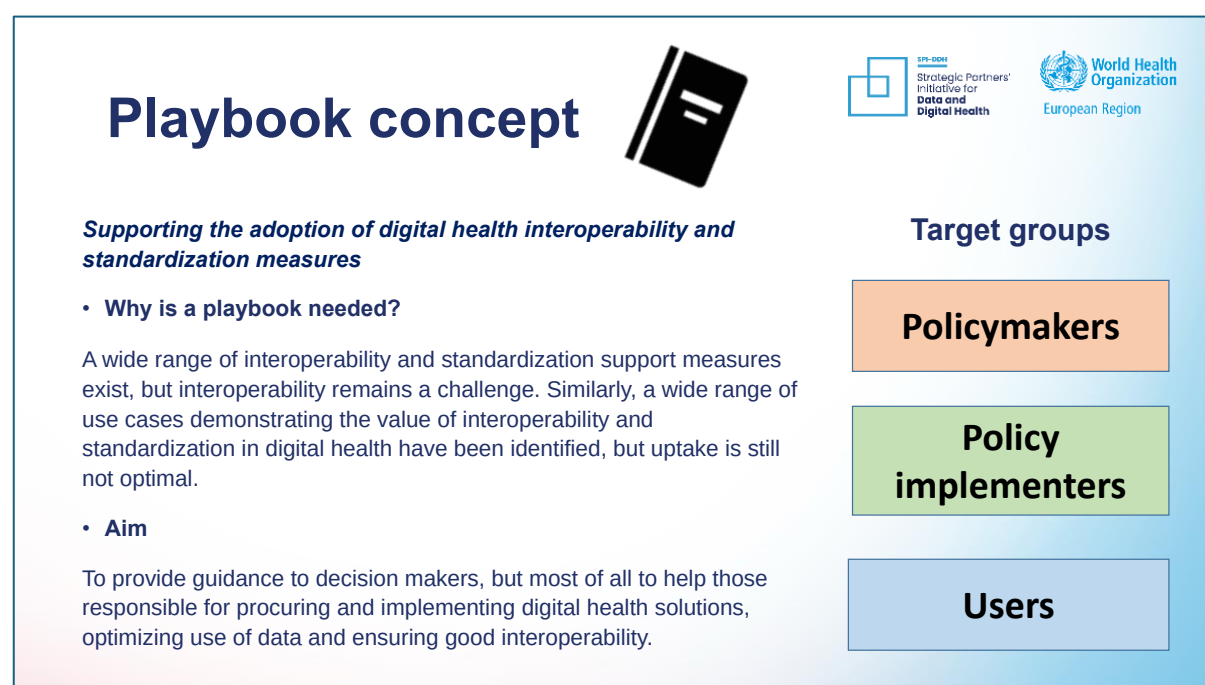
Focus: orchestrating interoperability within a national digital health ecosystem

Proposed output: an interoperability playbook

As health care systems become more digitalized, they deploy an increasingly wide range of digital solutions operating across multiple health and care providers; they also increasingly work across organizational and national borders and engage more actively with patients and informal carers. Ensuring patient safety in this evolution creates increasingly complex demands for the interoperability of digital health solutions and the data they use, both to provide care (primary use of data) and in the reuse of data collected for care provision in policy-making and research (secondary use).

Nongovernmental, governmental and industry bodies have developed a wealth of technical, semantic, organizational and legal standards, but their use is not always optimal. The working group's objective was to collect a wide range of demonstration narratives from implementers of digital health solutions that describe what interoperability challenges were encountered and whether they were overcome in the deployment of digital health solutions across a range of health and care settings. Based on the reports of successful deployments, as well as those that were abandoned, the group would develop a playbook designed to act as an orchestration guide for organizations intending to deploy new digital health solutions, or change existing ones, to help them identify which standards and other tools are best used to address the challenges of interoperability. The playbook would provide an approach for interoperability orchestration, an inventory of resources and a series of demonstration narratives (Fig. 3).

Fig. 3. The concept of the interoperability playbook

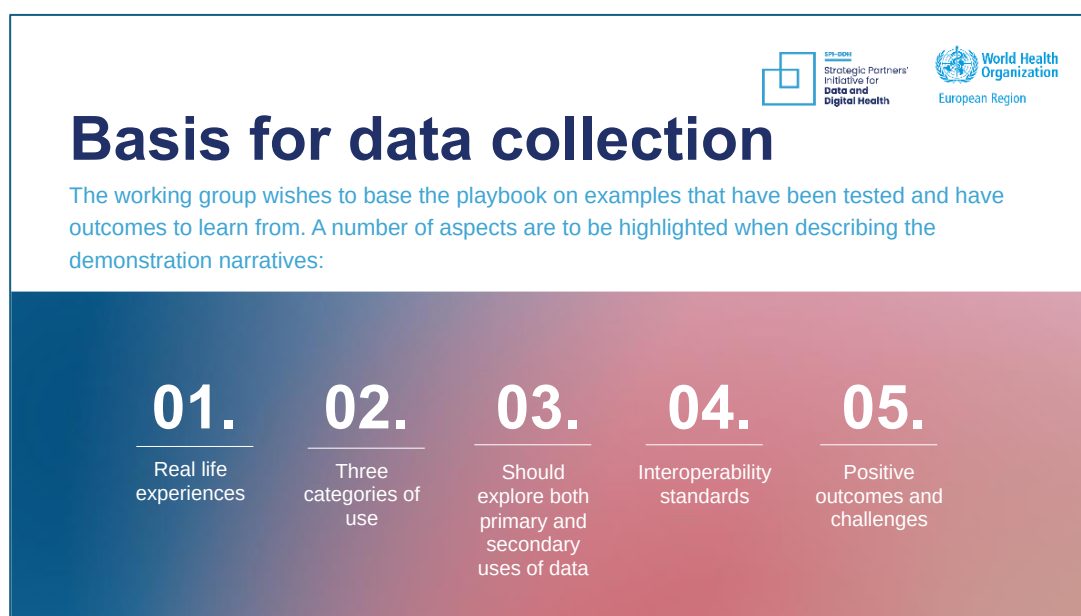


Building on the guideline for interoperability set out in the Refined eHealth European Interoperability Framework (5), a web-based tool has been developed to allow working-group partners, as well as other SPI-DDH members, to populate the demonstration narrative template.

It includes free text elements in which contributors are invited to describe the health and care problem that the digital health solution was designed to address, setting out the organizations and professionals that used the solution, and how it contributed to patient experience of the care provided. It also includes a list of a wide range of existing technical and semantic standards, and asks contributors to indicate which were included in the procurement terms of reference and which were then deployed.

Once the demonstration narratives are collected, they will serve as the basis of drafting the orchestration guidelines for policy-makers, policy executors and users and will also help to refine the list of interoperability resources that will make up the full interoperability playbook (Fig. 4 and 5).

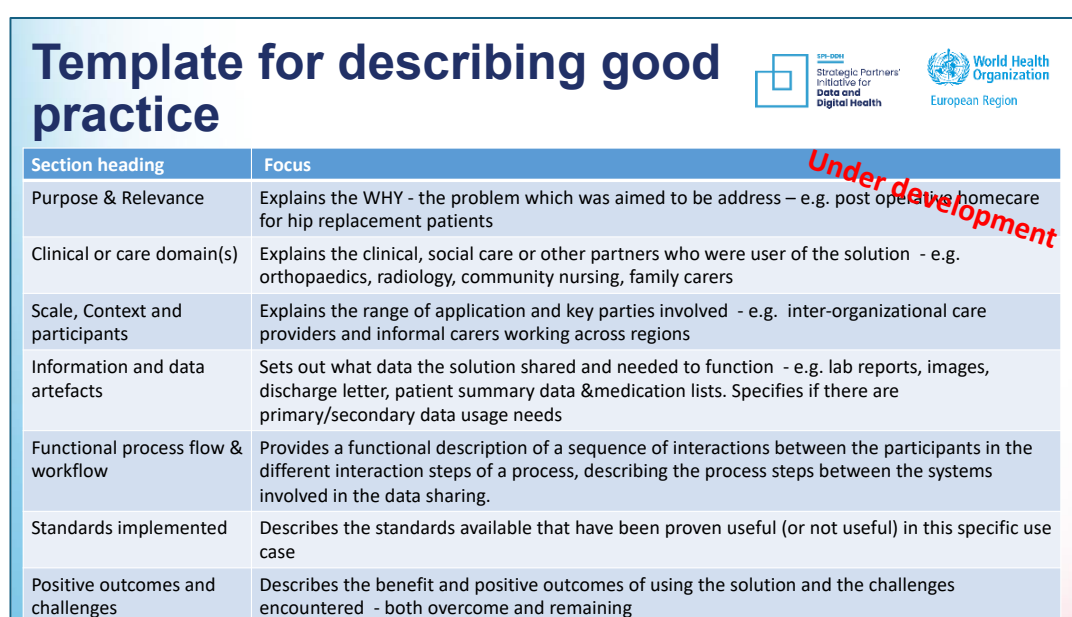
Fig. 4. Basis for data collection for working group 3



The slide titled "Basis for data collection" features logos for the Strategic Partners' Initiative for Data and Digital Health, the World Health Organization, and the European Region. The main text states: "The working group wishes to base the playbook on examples that have been tested and have outcomes to learn from. A number of aspects are to be highlighted when describing the demonstration narratives:". Below this, five numbered points are listed on a gradient background:

- 01.** Real life experiences
- 02.** Three categories of use
- 03.** Should explore both primary and secondary uses of data
- 04.** Interoperability standards
- 05.** Positive outcomes and challenges

Fig. 5. Template for describing good practice



The table titled "Template for describing good practice" includes logos for the Strategic Partners' Initiative for Data and Digital Health, the World Health Organization, and the European Region. A red diagonal watermark reading "Under development" is overlaid on the table. The table structure is as follows:

Section heading	Focus
Purpose & Relevance	Explains the WHY - the problem which was aimed to be address – e.g. post operative homecare for hip replacement patients
Clinical or care domain(s)	Explains the clinical, social care or other partners who were user of the solution - e.g. orthopaedics, radiology, community nursing, family carers
Scale, Context and participants	Explains the range of application and key parties involved - e.g. inter-organizational care providers and informal carers working across regions
Information and data artefacts	Sets out what data the solution shared and needed to function - e.g. lab reports, images, discharge letter, patient summary data & medication lists. Specifies if there are primary/secondary data usage needs
Functional process flow & workflow	Provides a functional description of a sequence of interactions between the participants in the different interaction steps of a process, describing the process steps between the systems involved in the data sharing.
Standards implemented	Describes the standards available that have been proven useful (or not useful) in this specific use case
Positive outcomes and challenges	Describes the benefit and positive outcomes of using the solution and the challenges encountered - both overcome and remaining

Public health working group. Addressing health-care workforce mental health through a data-driven, digital lens for resilience and operational efficiency

Focus: prioritizing the mental health of the health care workforce, particularly addressing burnout through data-driven, digital health solutions that improve well-being and health care system resilience, while balancing technological innovation and operational efficiency

Proposed output: work in progress (actual solution not yet defined): a solution that should employ data and digital technologies to empower health care professionals to prevent/monitor/treat burnout, by optimizing health care workload and improving operational efficiency

The working group approached the topic by prioritizing specific mental health challenges in the health care workforce, especially in relation to reducing burnout and short-term leave in secondary and tertiary care. Stakeholders from both the public and private sectors expressed concerns about balancing evidence-based digital health solutions with operational efficiency and workload reduction. Feedback informed the working group's focus on specific problem statements that reflect a consensus on this theme (Box 3).

Box 3. Data fragmentation

"Fragmented data across health care systems hinders the efficient utilization of resources at both health care and primary care levels, limiting the support available for frontline health care workers. Without robust data integration and harmonization, critical insights are lost, resulting in inefficiencies that can impact the mental health and well-being of those on the frontlines. Addressing these gaps is essential for optimizing resource allocation and, improving the mental health support provided to health care professionals."

Group member, Working Group three

A mind map and a framing tree analysis developed using electronic whiteboard software were used to complete this assignment as well as the next one: identifying the root cause of the problem, and prioritizing the unmet need that the group wants to address. This multifaceted breakdown is set to help align priorities with Member States' interests and provide a solid foundation for practical, scalable solutions.

With the mental health of the health care workforce identified as the highest priority, the group agreed on refining one final problem statement and delving into the proof of concept through a two-tier process: by fostering a debate led by the representatives of Member States present (France, Spain and the United Kingdom) and by conducting literature screening.

Mental health issues – including burnout, anxiety and depression – increasingly affect health care professionals and caregivers in general. These in turn affect health care professionals' well-being, the quality of care they provide and the stability of health care systems. It is therefore crucial to take a proactive and data-driven approach to effectively address professionals' mental health needs, thereby improving individual well-being, service provision and institutional resilience.

Various group members interpreted the results of the rising rates of burnout among health care professionals as encompassing not only workforce shortages but also violence, poor working conditions and work–life balance, low quality of care and patient safety, and long working hours, leading to organizational inefficiencies and higher health care expenditure.

Subsequently, the group will focus on exploring how digital technologies can reduce the workload of health care professionals while incorporating interconnected themes such as resource allocation and managerial support.

Key areas to be explored and validated before the solution ideation exercise, to define the group's approach of the desired output, include the following:

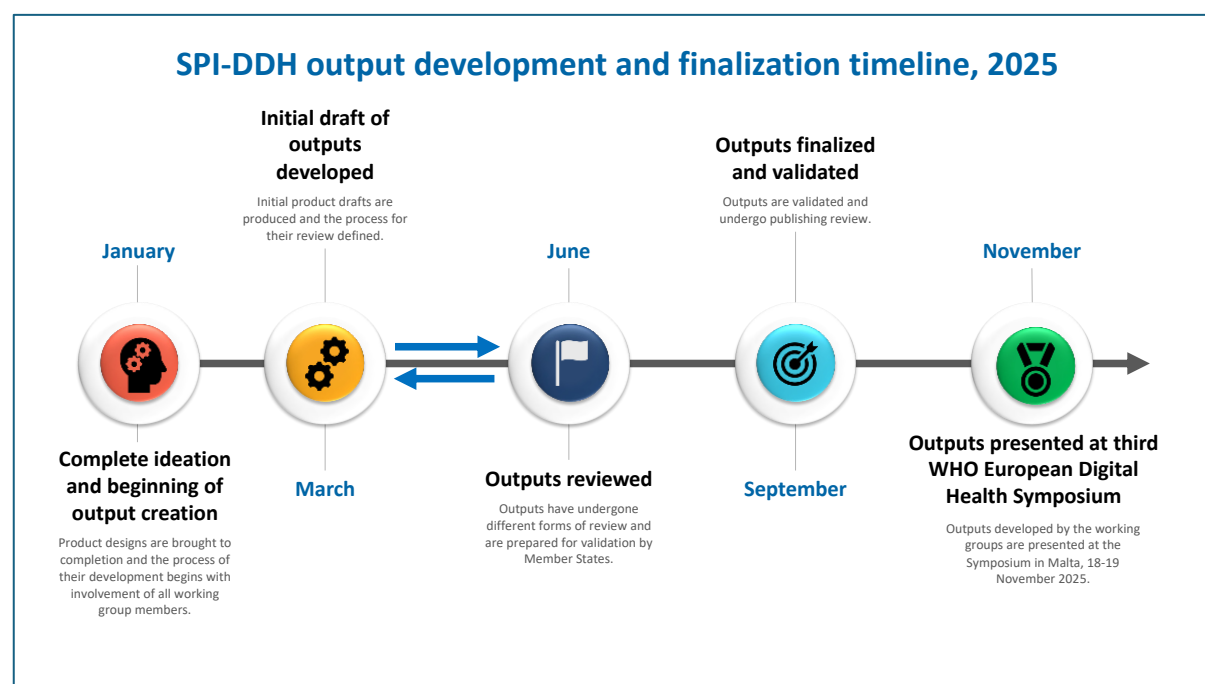
1. digital tools to optimize workflows:
 - clinical decision support systems
 - process automation and skills augmentation
2. technologies to empower patients:
 - tools to reduce unnecessary clinical visits
 - solutions to improve patient self-management and health literacy
3. resource allocation and managerial support:
 - evidence on effective resource optimization in health care settings
 - digital tools supporting managerial oversight and staff well-being.

Looking ahead – on track to deliver for countries

Over the first half of 2025, the SPI-DDH working groups, supported by the WHO Regional Office for Europe Secretariat, will transition from ideation to product scoping, development and refinement. This process will engage all members to provide content in developing the products outlined above.

Fig. 6 depicts a timeline of milestones for the development and completion of the working groups' outputs.

Fig. 6. SPI-DDH timeline for development and completion of outputs



The working groups will also be invited to deliver additional awareness-raising activities and provide innovative methods to support Member States through such activities as creating knowledge communities, video spotlights and living case study illustrations.

Aligning the work of SPI-DDH with other key international initiatives for data and digital health

Further, SPI-DDH will seek to align its work with that of other selected initiatives of relevance to the pan-European agenda for health systems transformation through use of data and digital health technologies. In particular, SPI-DDH activities will be closely aligned with and contribute to those of the digital health and innovation subgroup of the European Health Information Network. The Network was established to guide WHO's work under the action funded by the European Union (EU) to support Member States in the WHO European Region in strengthening health information systems and boosting health data governance.

Other efforts will also be made to align the outputs of the SPI-DDH working groups with major topic-specific and policy developments/projects in the WHO European Region, such as the European Health Data Space, the Digital Agenda for Europe, Europe's Digital Decade 2030, EU laws (the Artificial Intelligence, Digital Services and Digital Markets acts), the EU regulations on medical devices and in vitro diagnostic medical devices, the European Medicines Agency and the European Electronic Health Record Exchange Format.

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¹ All references were accessed 21 December 2024

The WHO Regional Office for Europe

The World Health Organization (WHO) is a specialized agency of the United Nations created in 1948 with the primary responsibility for international health matters and public health. The WHO Regional Office for Europe is one of six regional offices throughout the world, each with its own programme geared to the particular health conditions of the countries it serves.

Member States

Albania	Greece	Portugal
Andorra	Hungary	Republic of Moldova
Armenia	Iceland	Romania
Austria	Ireland	Russian Federation
Azerbaijan	Israel	San Marino
Belarus	Italy	Serbia
Belgium	Kazakhstan	Slovakia
Bosnia and Herzegovina	Kyrgyzstan	Slovenia
Bulgaria	Latvia	Spain
Croatia	Lithuania	Sweden
Cyprus	Luxembourg	Switzerland
Czechia	Malta	Tajikistan
Denmark	Monaco	Türkiye
Estonia	Montenegro	Turkmenistan
Finland	Netherlands (Kingdom of the)	Ukraine
France	North Macedonia	United Kingdom
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Germany	Poland	

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